INCH-POUND
MIL-PRF-45333F(AT)
14 October 1997
SUPERSEDING
MIL-PRF-45333E(AT)
29 November 1996

PERFORMANCE SPECIFICATION

TRAILER, FLAT-BED: 10 TON, 4 WHEEL M345

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a tactical flat-bed, 10 ton, 4 wheel type trailer designated as M345.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Other Government documents, drawings, and publications. The following other Governments documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

AMSC N/A FSC 2330

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NUCLEAR REGULATORY COMMISSION (NRC)

Code of Federal Regulations (CFR) - Title 10, Parts 30 and 40.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available form the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Order of precedence. In the event of a conflict between the text of this document and references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified (see 6.2), the design, materials, and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, ownership and support, and operating environment requirements specified. Recycled, recovered, or environmentally preferable materials shall be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Operating requirements. Each trailer shall provide the following functional, operational, and performance capabilities.

3.3.1 Seals. Seals shall prevent entrance of foreign matter into the bearings. All bearings seals shall restrict the leaking of lubricants from bearings.

3.3.2 Electrical system. Electrical circuits shall maintain continuity from end to end without evidence of internal or external shorts during all trailer operating conditions. All stop, signal, and breaking lights shall be of the standard commercial lighting for trailers.

3.3.3 Tires and inner tubes. Tires shall be cross country tires with tire flaps. Inner tubes shall be of the proper type and size for trailer applications.

3.3.4 Adjustable mechanisms. All adjustments mechanisms shall function properly and maintain adjustments settings during all vehicle operating conditions.

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3.3.5 Highway (paved) and gravel roads, and speeds. The trailer shall operate on highway and gravel roads without damage to the trailer or tow vehicle.

3.3.6 Cross-country payload and speeds. The trailer shall be capable of being towed over cross country terrain without damage to the trailer or tow vehicle.

3.3.7 Grade and slope operation. The trailer shall be capable of ascending and descending longitudinal grades and side slopes without slipping or upsetting.

3.3.8 Service brakes. The combined service brakes of the trailer and the tow vehicle shall bring the combination to a positive stop.

3.3.8.1 Service breaking system. The service breaking system shall not display fluid leakage and shall maintain air pressure when applied to the wheels of the trailer.

3.3.9 Automatic actuation. The trailer shall be equipped with an automatic actuating device to apply the trailer brakes upon breakaway from the tow vehicle.

3.3.10 Fording ability. The vehicle and its components shall sustain no damage when towed through fresh or salt water.

3.3.11 Landing legs. Each landing leg shall be capable of raising, supporting, holding, and retracting its load without damage.

3.3.12 Turning ability. The trailer shall follow a turning tow vehicle without cramping, damage or interference.

3.3.13 Wheel and tire balancing. Each wheel assembly, tube and tire shall be balanced to preclude wheel tramp and drumming.

3.4 Interface requirements.

3.4.1 Design and construction. Trailer components, subassemblies and assemblies shall be fabricated and assembled into a complete trailer in accordance with the manufacturer’s design and construction (see 6.3).

3.5 Ownership and support requirements.

3.5.1 Reliability. The design goal of reliability of the M345 shall not be less than 6000 Mean Miles Between Failure (MMBF) of 45 percent (%) hard-surface roads, 50% secondary roads and 5% cross country operation.

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3.5.2 Maintainability. Total maintenance, excluding drivers/crew checks and services, shall not exceed 15 work-hours during 6000 miles (mi) of operation. This equates to a maintenance ratio of 0.075 at 30 operational mi, equivalent to one operational hour. The schedule maintenance interval shall be 3 months or 3000 mi, whichever comes first.

3.5.3 Safety. All exposed parts which are energized electrically shall be located, insulated, fully enclosed or guarded to prevent hazards to operating personnel and equipment functioning. All moving parts which are of such nature, or so located, as to be a hazard to operating or maintenance personnel shall be enclosed or guarded. Protective devices shall not impair operating functions.

3.5.3.1 Hazardous materials. The trailer shall not produce any hazards to personnel or the environment resulting from the use of asbestos, cadmium, or radioactive materials.

3.5.4 Painting and treating. Unless otherwise specified (see 6.2), the trailer shall be cleaned, treated, and painted in accordance with normal commercial practice to assure complete coverage and durability of the finish. Unless otherwise specified (see 6.2), the color shall be lusterless forest green.

3.5.5 Corrosion resistance. Materials and special treatments shall be used to prevent corrosion and to resist the harmful effects of dust, water, and fungi.

3.5.6 Identification and marking. Identification and marking shall be permanent and legible and shall be made of a material suitable to withstand the environmental requirements specified herein. The trailer shall be marked with the National Stock Number (NSN) and the manufacturer's name and part number.

3.6 Operating environment requirements.

3.6.1 Environment. The trailer shall operate under extreme conditions of weather and altitude and in ambient air temperatures ranging from -65 to 115 degrees Fahrenheit (°F). The complete trailer when in storage shall withstand a temperature of -80 °F without deterioration that may cause failure of any component.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Conformance inspection (see 4.1.2).

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4.1.1 First article inspection. When specified (see 6.2), first article inspection shall include all the verifications listed in table I (see 6.4).

4.1.2 Conformance inspection. Conformance inspection shall include the examinations and tests listed in table I as defined by the contract (see 6.5).

Table I. Verification Methods.

Title	Require-ment	Verifica-tion	Title	Require-ment	Verifica-tion
Operating requirements	3.3	4.2.2.1	Turning ability	3.3.12	4.2.2.11
Seals	3.3.1	4.2.2.2	Wheel, tire balancing	3.3.13	4.2.2.12
Electrical system	3.3.2	4.2.2.3	Interface requirements	3.4	4.2.3
Tires and inner tubes	3.3.3	4.2.2.4	Design and construction	3.4.1	4.2.3.1
Adjustable mechanisms	3.3.4	4.2.2.5	Ownership and support	3.5	4.2.4
Highway, gravel, speeds	3.3.5	4.2.2.5	Reliability	3.5.1	4.2.4.1
Cross-country, speeds	3.3.6	4.2.2.6	Maintainability	3.5.2	4.2.4.2
Grade and slope	3.3.7	4.2.2.7	Safety	3.5.3	4.2.4.3
Service brakes	3.3.8	4.2.2.8	Hazardous materials	3.5.3.1	4.2.4.3
Service braking system	3.3.8.1	4.2.2.8.1	Painting	3.5.4	4.2.4.4
Automatic actuation	3.3.9	4.2.2.8.2	Corrosion resistance	3.5.5	4.2.4.5
Fording ability	3.3.10	4.2.2.9	Identification, marking	3.5.6	4.2.4.4
Landing legs	3.3.11	4.2.2.10	Environment	3.6.1	4.2.5

4.2 Verification methods. Acceptable verification methods included in this section are visual inspection, measurement, sample tests, full-scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, similarity to previously-approved or previously-qualified designs, and inspection of contractor's records providing proof or certification that design, construction, processing, and materials conform to requirements. Applicable records shall include drawings, specifications, design data, receiving inspection records, processing and quality control standards, vendor catalogs and certifications, industry standards, test reports, and rating data.

4.2.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques, or equipment, including the application of statistical process control, tool control, or cost effective sampling procedures to verify performance. See the contract for alternatives that replace verification methods required by this specification.

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4.2.2 Operating requirements verification.

4.2.2.1 Seals. Trailer shall ford water (see 3.3.10) and be operated in mud, sand, and snow. After specified operations, trailer shall be checked for foreign matter in the bearings that are exposed to contamination during these operations and shall show no evidence of lubricant leakage.

4.2.2.2 Electrical system. Trailer shall be operated throughout all operating conditions and examined for functional requirements.

4.2.2.3 Tires and inner tubes. Tires and inner tubes shall be examined for functional requirements.

4.2.2.4 Adjustable mechanisms. Vehicle shall be operated throughout all operating conditions and examined for the proper function of adjustment mechanisms.

4.2.2.5 Highway. The trailer shall be loaded to a 26 000 pound (lb) payload, or with a simulation load of equal weight, uniformly distributed over the load area and shall be towed at an average speed of 50 miles per hour (mph) over relatively level, smooth, improved and pre-paved hand-surfaced roads, and a speed of 20 mph over gravel roads. The trailer shall track the tow vehicle in a straight line without deviating more than 3 inches to either side of the path of the tow vehicle and without exhibiting damage to the trailer or the tow vehicle.

4.2.2.6 Cross country. The trailer, loaded with cross country rated payload of 20 000 lb uniformly distributed over the load area, shall be operated on unimproved roads, trails, open fields, rolling hills and cross country terrain at average speeds of 20 mph without exhibiting damage to the trailer or the tow vehicle.

4.2.2.7 Grade and slope operation. The trailer and tow vehicle shall be operated on longitudinal grades up to 40 % and side slopes up to 20% and shall ascend and descend without slipping or upsetting.

4.2.2.8 Service brake. The trailer and tow vehicle shall be operated at a speed of 20 mph on dry, smooth and level road, free of loose material, and the service brakes applied. The vehicle shall come to a positive stop within 30 feet of service brake application.

4.2.2.8.1 Service brake system. The system shall be charged with air from 100 to 105 pounds per square inch (psi) and the brake air supply disconnected. After disconnection, service brakes shall be applied to all wheels of the trailer, not display fluid leakage, and maintain air pressure with a drop of no less than 0.5 psi per min and no more than 5 psi per min.

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4.2.2.8.2 Automatic actuation. The trailer shall be loaded to a 26 000 lb highway payload and placed on a 20% grade. The automatic actuating device shall maintain application and hold the trailer stationary for not less than 15 minutes (min).

4.2.2.9 Fording. The trailer shall be towed through fresh and salt water at a depth to entirely submerge the chassis for periods up to 15 min duration. Without the use of special kits and with a minimum of preparation or servicing before and after fording operations the vehicle and its components shall display no damage.

4.2.2.10 Landing leg. The trainer shall be loaded to a 26 000 lb highway payload and to a 20 000 lb cross country pay load and each landing leg shall be raised, support and hold its portion of the load, and retract without sustaining damage. When retracted in cross country operation, the vehicle shall be operated on unimproved roads, trails, open fields, and rolling hills without the landing legs hitting the ground.

4.2.2.11 Turning ability. The trailer and the tow vehicle shall be coupled and operated in a minimal turning circle without evidence of cramping, damage, or interference between the trailer and tow vehicle.

4.2.2.12 Wheel and tire balance. Each assembly of wheel, tube and tire shall be balanced and the vehicle operated at road speeds up to 50 mph without exhibiting unusual noise.

4.2.3 Interface requirements verification.

4.2.3.1 Design and construction. The trailer shall be examined to assure proper design and construction requirements.

4.2.4 Ownership and support requirements verification.

4.2.4.1 Reliability. Trailer reliability shall be verified at a point estimate value equal to or greater than 6000 mi which shall be demonstrated during the road test as specified in table II. For calculation of the MMBF, a failure shall be defined as a condition which prevents operation, or reduces performance below essential levels, or indicates to the crew that further operation would be unsafe, and might result in extensive damage to the equipment. Any of the foregoing conditions which cannot be corrected by the crew in 30 min, using the tools land parts normally carried by the M345 or the tow vehicle, constitutes failure. Maintenance and human induced failure are excluded.

4.2.4.2 Maintainability. A maintenance ratio of less than, or equal to, specified requirement shall be demonstrated during the road test in table II.

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Table II. 6000 mile road test.

Course	Mileage and speeds
Hard surfaced roads	2400 mi at varying speeds up to maximum.
Gravel and dirt roads	1500 mi at varying speeds up to maximum.
Level cross-country	1380 mi at varying speeds up to maximum.
Belgian block 1/	120 mi at varying speeds up to maximum
Hilly cross-country	600 mi at speeds applicable to trailer characteristics.

1/ = The 6000 mi road test shall include 50 mi of gravel and 30 mi of belgian block without payload at start and completion of highway and cross-country operations, for a total of 160 mi.

4.2.4.3 Safety. The trailer shall be examined to assure that hazards to operating and maintenance personnel have been eliminated.

4.2.4.4 Painting and identification. Trailer shall be examined to assure that painting and marking is as specified.

4.2.4.5 Corrosion resistance. Trailer shall be examined to assure that corrosion resistance is as specified.

4.2.5 Environment. The trailer shall be operated and stored at specified temperatures and checked for functional requirements and evidence of deterioration.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general and explanatory nature that may be helpful, but is not mandatory.)

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6.1 Intended use. Trailers covered by this specification are intended for use in hauling items of construction engineering equipment or similar applications during tactical military operations, and for use under extreme or unusual conditions of climate, weather, and terrain.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).
c. When first article inspection is required (see 3.1 and 4.1.1).
d. If design, materials, or manufacturing are other than as specified (see 3.2).
e. If treatment and painting are other than as specified (see 3.5.4).
f. Packaging requirements (see 5.1).

6.3 Supplemental drawings. Historically, trailer components, subassemblies and assemblies have been fabricated and assembled into a complete trailer in accordance with Drawing 8358914; Trailer, Flatbed: 10 Ton, 4 Wheel, M345. Copies of these drawings are available from the U.S. Army Tank-automotive and Armaments Command, AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

6.4 First article. When requiring a first article inspection, contracting documents should provide specific guidance to offerors. This guidance should cover whether the first article is a first article sample, a first production item, or the number of test items. These documents should also include specific instructions regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Pre-solicitation documents should provide Government waiver rights for samples for first article inspection to bidders offering a previously acquired or tested product. Bidders offering such products who wish to rely on such production testing must furnish evidence with the bid that prior Government approval is appropriate for the pending contract.

6.5 Conformance inspection. Affordable conformance inspection with confidence varies depending upon a number of procurement risk factors. Some of these factors include: Contractor past performance, government schedules and budget, product material and design maturity, manufacturing capital equipment and processes applied, the controlled uniformity of those processes, labor skill and training, and the uniformity of measuring processes and techniques. During the solicitation, contracting documents should indicate those tests desired from table I and their designated frequency based on a risk assessment for the procurement.

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6.6 Subject term (key word) listing.

Engineering equipment
Hauling
Recovery
Transporter
Tractors

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Army - AT

Preparing Activity:
Army - AT

(Project 2330-0302)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-45333F(AT)	**2. DOCUMENT DATE *(YYMMDD)*** 971014

3. DOCUMENT TITLE

TRAILER, FLAT-BED: 10 TON, 4 WHEEL M345

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	**b. ORGANIZATION**	
c. ADDRESS *(Include Zip Code)*	**d. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** ***(If applicable)***	**7. DATE SUBMITTED *(YYMMDD)***

8. PREPARING ACTIVITY

a. NAME	**b. TELEPHONE *(Include Area Code)*** **(1) Commercial** **(2) AUTOVON** **(810) 574-8745** **786-8745**
c. ADDRESS *(Include Zip Code)* **Commander** **U.S. Army Tank-automotive and Armaments Command** **ATTN: AMSTA-TR-E/BLUE** **Warren, MI 48397-5000**	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** **Defense Quality and Standardization Office** **5203 Leesburg Pike, Suite 1403** **Falls Church, VA 22041-3466** **Telephone (703) 756-2340 AUTOVON 289-2340**

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